

Is Flint's water “fine on average”?

Lead level in 71 Flint home water samples, Jan–Jun 2015 (parts per billion)

EXAMPLE POSTER · Data · Graph · Sentence

The claim we judged: *“The average home is at 7.31 ppb, under the 15 ppb action level, so the water is fine.”*

DATA

Individuals: 71 water samples, one per Flint home.

Variable: lead level (ppb), quantitative.

Minimum 0 ppb

Q1 2 ppb

Median 3 ppb

Q3 7 ppb

Maximum 104 ppb

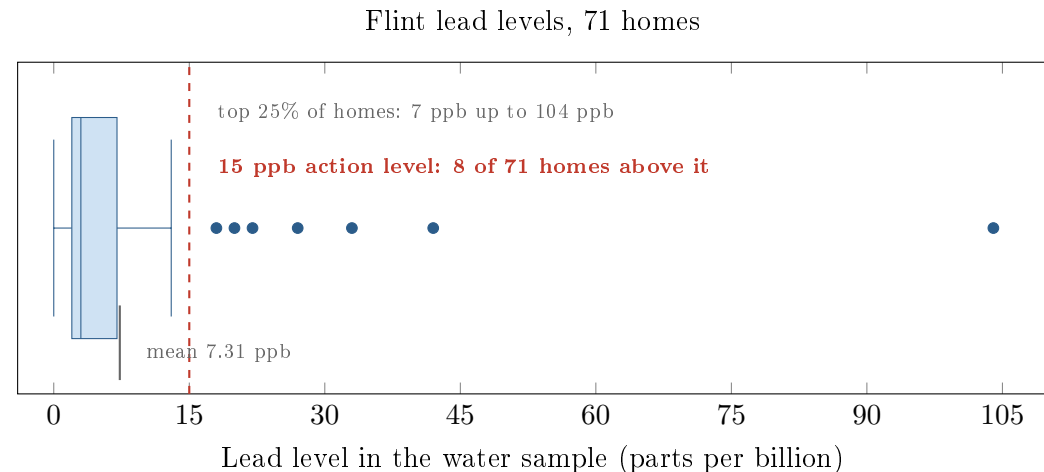
Mean 7.31 ppb

IQR $7 - 2 = 5$ ppb

Upper fence: $7 + 1.5(5) = 14.5$ ppb.
The right whisker stops at 13; every value above 14.5 is an outlier.

Above 15 ppb: 8 of 71 = 11.3%.

GRAPH



Each of the four pieces of the box plot holds about 25% of the 71 homes: 0 to 2, 2 to 3, 3 to 7, and 7 to 104 ppb. The whisker stops at 13, the last value under the 14.5 fence; the seven outlier dots are placed at illustrative values because the exact readings are not printed in the lesson. The five-number summary and 8 of 71 above 15 ppb are from Lessons 1.7 and 1.8.

SENTENCE

Half of the 71 Flint homes tested at or below 3 ppb of lead, but the top quarter ran from 7 ppb all the way to 104 ppb, and 8 homes (11.3%) were over the 15 ppb action level. The mean of 7.31 ppb is more than double the median because those few dangerous homes drag it up, so “the average home is fine” hides exactly the homes that are not. Report the median and IQR (3 ppb, 5 ppb) and the 11.3%, not the mean.

Talk-track for the example poster

AP Statistics · Unit 1 Poster · Board demo

[SPEAK] WHAT TO SAY AT THE BOARD (about 8 minutes)

1. **Read the claim out loud.** “The average home is at 7.31 ppb, under 15, so the water is fine.” Ask: what would you need to see before you believed that? Take two answers. Do not evaluate them yet.
2. **DATA box.** Point out it opens with WHO (71 homes) and WHAT (lead level, ppb). “Every number on every poster in this room needs those two things or it is not a statistic yet, it is a number.” Then the five-number summary. Ask: which is bigger, the mean or the median? By how much? Which direction is the tail?
3. **GRAPH box.** Build the box plot with the class: box from 2 to 7, median line at 3, whisker to 0, right whisker to 13 because 14.5 is the fence. “Where do the other seven homes go?” Dots. Draw the 15 ppb line. “How many homes are to the right of that line?” Eight, 11.3%. Then the 25% labels: “each piece of a box plot is a quarter of the homes, no matter how long it looks.”
4. **SENTENCE box.** Read it. Then ask the class to find, in the sentence: the context (Flint homes, lead, ppb), the comparative numbers (3 vs 7.31, 8 of 71), the judgment (“hides exactly the homes that are not”), and the recommendation (median and IQR). “That is what your two or three sentences must do: context, numbers from the graph, a judgment, and a recommendation.”
5. **Hand out the sheets.** “Your sheet has a claim somebody made. Solve (a) through (d) together before anyone touches the poster board. Your poster is these three boxes: Data, Graph, Sentence.”

[WARN] MISTAKES TO NAME WHILE YOU DEMO (these are the class's own)

- “7.31 is the typical home.” The mean follows the tail; the median (3) does not. Period E was flagged on this across 12 answers.
- “The whisker goes to 104.” Whiskers stop at the last non-outlier (13); outliers are separate dots.
- “75% of homes are above the median.” Each section is 25%; above the median is 50%.
- “11.3%” with no sentence around it. Who, what, units, and the 15 ppb question are what make it mean something.